

ANSWER KEY — Integration Lesson 35 Test: Volumes II — Off-Axis & Cross-Sections

For the grader · full working shown · give credit for a correct setup with small arithmetic slips · setup-only questions earn full credit with no evaluation

1. (a) $y = \sqrt{x}$ revolved about $y = -3$: radius at x ? (b) About $y = 6$ (curve below): radius?

Work: radius = distance from the spin line to the curve. (a) Line below curve: $r = \sqrt{x} - (-3) = \sqrt{x} + 3$. (b) Line above curve: $r = 6 - \sqrt{x}$. **Answer: (a) $r = \sqrt{x} + 3$ (b) $r = 6 - \sqrt{x}$.**

2. Region between $y = x$ and $y = -1$, $0 \leq x \leq 3$, revolved about $y = -1$. Volume?

Work: region touches the spin line $\rightarrow$ full disks, radius = $x + 1$. $V = \pi \int_0^3 (x + 1)^2 dx = \pi[(x + 1)^3/3]_0^3 = \pi(64/3 - 1/3) = 63\pi/3$. **Answer: $V = 21\pi$.**

3. Region between $y = \sqrt{x}$ and $y = -2$, $0 \leq x \leq 1$, revolved about $y = -2$. Volume?

Work: radius = $\sqrt{x} + 2$. Expand: $(\sqrt{x} + 2)^2 = x + 4\sqrt{x} + 4$ (don't drop the middle term!). $V = \pi \int_0^1 (x + 4\sqrt{x} + 4) dx = \pi[x^2/2 + (8/3)x^{3/2} + 4x]_0^1 = \pi(1/2 + 8/3 + 4) = \pi(3/6 + 16/6 + 24/6)$. **Answer: $V = 43\pi/6$.**

4. Region bounded by $x = y^2$, the y -axis, and $y = 3$, revolved about the y -axis. Volume?

Work: horizontal slices: disk radius $x(y) = y^2$. $V = \pi \int_0^3 (y^2)^2 dy = \pi \int_0^3 y^4 dy = \pi[y^5/5]_0^3 = \pi \cdot 243/5$. **Answer: $V = 243\pi/5$.**

5. Base between $y = x$ and the x -axis on $[0, 3]$; square cross-sections. Volume?

Work: footprint $w(x) = x$, slice area $A = w^2 = x^2$ (NO π — squares aren't circles). $V = \int_0^3 x^2 dx = [x^3/3]_0^3 = 27/3$. **Answer: $V = 9$.** (Writing 9π loses the point.)

6. Base between $y = \sqrt{x}$ and the x -axis on $[0, 2]$; semicircle cross-sections (diameter on base). Volume?

Work: diameter $w = \sqrt{x} \rightarrow$ radius $w/2 \rightarrow A = \frac{1}{2}\pi(w/2)^2 = (\pi/8)w^2 = (\pi/8)x$. $V = (\pi/8) \int_0^2 x dx = (\pi/8)[x^2/2]_0^2 = (\pi/8)(2)$. **Answer: $V = \pi/4$.**

7. Region bounded by $y = x$, $y = 3$, and the y -axis, revolved about $y = 3$. Volume + cone check.

Work: spin line above the region: radius = $3 - x$, from $x = 0$ to 3 . $V = \pi \int_0^3 (3 - x)^2 dx = \pi[-(3 - x)^3/3]_0^3 = \pi(0 - (-27/3)) = 9\pi$. Cone check: radius 3, height 3 $\rightarrow \frac{1}{3}\pi(3^2)(3) = 9\pi$ ✓. **Answer: $V = 9\pi$.**

8. Region between $y = \sqrt{x}$ and the x-axis on $[0, 4]$, revolved about $y = -2$. Set up only.

Work: the region floats 2 units above the spin line $\rightarrow$ washers. Outer radius $= \sqrt{x} + 2$ (curve to line); inner radius $= 0 - (-2) = 2$ (x-axis to line). **Answer:** $V = \pi \int_0^4 [(\sqrt{x} + 2)^2 - 2^2] dx$. (No evaluation required. Common error: forgetting the inner radius 2 and writing a disk integral.)

9. Base between $y = 3$ and $y = x$ on $[0, 3]$; equilateral-triangle cross-sections. Volume?

Work: footprint $w(x) = 3 - x$; equilateral area $A = (\sqrt{3}/4)w^2$. $V = (\sqrt{3}/4) \int_0^3 (3 - x)^2 dx = (\sqrt{3}/4) [-(3 - x)^3/3]_0^3 = (\sqrt{3}/4)(27/3) = (\sqrt{3}/4)(9)$. **Answer:** $V = 9\sqrt{3}/4$.

10. Same base: Solid A (squares) has $V = 10$; Solid B has semicircles. (a) Volume of B? (b) Why no π in A?

Work: (a) Both volumes $= (\text{shape constant}) \cdot \int w^2 dx$ over the same base, so $\int w^2 dx = 10$ for squares (constant 1). Semicircles use constant $\pi/8$: $V_B = (\pi/8)(10)$. **Answer: (a) $V = 5\pi/4$. (b) π only enters a volume through the area of a circular slice (πr^2); a square's area w^2 has no π , so none can appear.**